

MYELOID NEOPLASIA

Targeting PRMT9 overcomes venetoclax resistance in AML by modulating splicing and inhibiting translation

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KEY POINTS

- PRMT9 promotes VEN resistance by facilitating aberrant RNA splicing and enhancing translational activity.
- Pharmacologic inhibition of PRMT9 synergizes with VEN to eliminate drug-resistant AML cells.

Arginine methylation catalyzed by protein arginine methyltransferases (PRMTs) is required for cancer cell proliferation, but whether PRMTs mediate resistance to therapy remains unclear. Here, we performed loss-of-function screens in venetoclax-resistant (VEN-R) acute myeloid leukemia (AML) patient-derived xenograft cells and found that PRMT9 plays a critical role in promoting VEN resistance. Specifically, VEN-R AML samples exhibited high levels of PRMT9, and PRMT9 inhibition resensitized AML cells to VEN treatment. In preclinical resistant models, genetic ablation of PRMT9 synergized with VEN to eradicate AML cells. Consistently, pharmacologic inhibition of PRMT9 combined with VEN produced similar effects in VEN-R AML mouse models. Mechanistically, PRMT9 ablation disrupted RNA splicing by inducing exon skipping in mRNA encoding ALG13, an uridine diphosphate (UDP)-N-acetylglucosaminyltransferase subunit, thereby down-regulating expression of the VEN efflux transporter encoded by the adenosine triphosphate-binding cassette subfamily C member 1 gene. PRMT9 inhibition also suppressed protein synthesis, leading to downregulation of short-lived oncoproteins such as MCL1. These findings establish a connection between PRMT9-mediated arginine methylation and poor VEN responsiveness and demonstrate that targeting PRMT9 may represent a viable strategy to overcome VEN resistance.

Introduction

The prognosis for patients with acute myeloid leukemia (AML) remains poor.¹ Although allogeneic stem cell transplantation represents a curative approach, its utility is limited because of the presence of comorbidities.² Venetoclax (VEN), an oral BCL-2 homology domain 3 (BH3) mimetic, was approved by the US Food and Drug Administration in 2018 for use in combination with hypomethylating agents, such as azacitidine (AZA), in order to treat older patients with AML ineligible for intensive chemotherapy.³ However, the therapeutic efficacy of the regimen is hindered by the short duration of responses and the emergence of drug resistance.³

Mechanisms of VEN resistance have been studied. Activating mutations in *RAS*, *PTPN11*, or *FLT3* may drive a dependency shift from BCL2 to BCL-XL or MCL1, resulting in poor responsiveness to VEN.³⁻⁷ Recently, newly acquired *BAX* mutations

have been identified in a subset of patients with AML.⁸ Furthermore, *TP53* aberrations, present in ~5% to 10% of newly diagnosed patients with AML, are associated with short response duration to VEN.^{9,10} However, genetic mutations are not always detected in most VEN-R patients.^{11,12} This underscores the need to investigate mutation-independent mechanisms contributing to VEN resistance. Accordingly, aberrantly active RNA translation is known to promote VEN resistance.¹³ Accordingly, treatment with homoharringtonine (HHT), a protein synthesis inhibitor, resensitized patients with AML resistant to VEN in clinical settings.¹³ Disruption of the splicing machinery has also emerged as a strategy to overcome VEN resistance.^{14,15}

Arginine methylation, a prevalent posttranslational modification catalyzed by protein arginine methyltransferases (PRMTs), plays a role in various cellular processes, including RNA splicing and translation.¹⁶ Accordingly, modulating PRMT

activity largely affects pre-mRNA splicing by altering the activity of splicing factors. PRMTs also catalyze methylation of ribosomal proteins, which is crucial for ribosome assembly and affects translational fidelity. Several recent studies report that PRMT inhibition affects alternative splicing and suppresses translation of targets involved in apoptotic sensitivity, such as MCL1 or BCL-XL.¹⁶⁻¹⁸ Thus, use of PRMT inhibitors may represent a promising approach.¹⁹ Nevertheless, a comprehensive investigation of how individual PRMTs modulate sensitivity to mitochondrial apoptosis in AML cells is lacking.

Here, we used genetic and pharmacological tools to ablate individual PRMTs in VEN-resistant (VEN-R) AML patient-derived xenograft (PDX) cells and a VEN-R cell line. Specifically, we used 2 complementary assays to assess apoptotic sensitivity. Among the PRMTs evaluated, PRMT9 emerged as an exploitable target. Accordingly, PRMT9 inhibition preferentially enhanced VEN responsiveness, and notably, VEN-R cells displayed higher PRMT9 levels than their VEN-sensitive counterparts. Moreover, in preclinical mouse models, targeting PRMT9 synergized with VEN treatment to eliminate AML cells. Mechanistically, we found that enhanced VEN sensitivity was associated with alterations in the splicing machinery, subsequent downregulation of glycosyltransferase activity, and repression of oncoprotein synthesis.

Methods

Patient samples

Deidentified, clinically annotated primary samples were collected from patients with AML at City of Hope Comprehensive Cancer Center, including specimens derived from peripheral blood (PB) or bone marrow (BM). Clinical annotations are provided in supplemental Table 1, available on the *Blood* website.

Mice

In all experiments, male and female mice aged 6 to 10 weeks were used, including Kmt2a^{tm2(MLLT3)Thr/KsyJ} (strain number 009079, MLL-AF9 knockin; The Jackson Laboratory), NOD.Cg-Prkdc^{scid} Il2rg^{tm1Wjl/SzJ} (strain number 005557, NSG; The Jackson Laboratory), and Prmt9^{fllox/fllox} (strain number S-CKO-00255; Cyagen Biosciences) mice. Mice were housed at the City of Hope Animal Resource Center. All animal care and experimental procedures were conducted in accordance with institutional guidelines and protocols approved by the Institutional Animal Care and Use Committee at the City of Hope National Medical Center. Mice were maintained on a 14-hour light/10-hour dark cycle, with temperatures controlled between 65 and 75°F and humidity maintained at 40% to 60%.

Statistics

Statistical analyses were performed using GraphPad Prism 10, and statistical significance was defined as $P < .05$. Data are presented as mean $\pm$ standard deviation unless otherwise specified. Statistical comparisons were performed using unpaired 2-tailed Student *t* tests, 2-way analysis of variance, and 1-way analysis of variance. (* $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$) The log-rank (Mantel-Cox) test was used to compare in vivo efficacy survival curves.

Results

A loss-of-function screen shows that PRMT9 inhibition resensitizes AML cells to VEN

We asked whether knockout (KO) of individual PRMTs would enhance VEN responsiveness. Our data-mining analysis of published CRISPR-KO screens treated with VEN-associated regimen²⁰⁻²² revealed synthetic lethal interactions between VEN and loss of PRMT1, PRMT4, PRMT5, PRMT6, PRMT7, or PRMT9 (Figure 1A; supplemental Figure 1A-B).

To determine whether loss of individual PRMTs promotes cellular sensitivity to mitochondrial apoptosis, we established VEN-R models using AML PDX (AML1 and AML2) as well as Molm13 cells. AML1-VEN-R and AML2-VEN-R PDX models were established by transplanting 1 million AML1 or AML2 blasts individually into NSGS mice. After successful engraftment, defined as $\sim 1\%$ circulating blasts in PB, mice were treated once daily with 100 mg/kg VEN by oral gavage 5 days a week. Once disease progression was observed, as indicated by an increase in circulating blasts to 20% in PB (Figure 1B), VEN-R human leukemic cells (CD45^{dim}/CD33⁺) were isolated from mouse BM for analysis. Although VEN resistance may originate from monocytic subsets,²³ both our resistant models (AML1 and AML2) showed no significant changes in expression of CD34 or monocytic markers (CD11B, CD14, or CD68) relative to respective parental cells (supplemental Figure 1C). Targeted sequencing⁸ of paired PDX models, together with newly added VEN-relapsed samples (AML3-5), revealed features consistent with commonly reported genotypes seen in cases with VEN resistance²⁴ (supplemental Figure 1D). Molm13 VEN-R cells were generated by exposing parental lines to increasing VEN doses over a 12-week period. Compared to corresponding parental controls, resistant models displayed notable VEN resistance based on elevated VEN 50% inhibitory concentration (IC₅₀) values (Figure 1C-D).

We then knocked down PRMTs individually in VEN-R models and assayed sensitivity to mitochondrial apoptosis using 2 approaches: an annexin V-based apoptosis assay and Bcl-2 interacting mediator of cell death (BIM)-based dynamic BH3 profiling.²⁵ We did not knock down PRMT8 in AML cells, as PRMT8 levels are undetectable in hematopoietic lineages.²⁶ PRMT9 knockdown (KD) significantly resensitized Molm13 VEN-R cells to VEN, based on increased apoptosis, reduced cell viability, and enhanced BH3 priming (Figure 1E-H; supplemental Figure 8A). To verify these results, we treated Molm13 VEN-R, AML1-VEN-R, and AML2-VEN-R cells with currently available PRMT pharmacological inhibitors MS023 (PRMT1, PRMT3, PRMT4, and PRMT6), SGC707 (PRMT3), EPZ025654 (CARM1), EPZ015666 (PRMT5), EPZ020411 (PRMT6), SGC3027 (PRMT7), and LD2 (PRMT9), respectively. PRMT inhibition was confirmed by remarkable suppression of methylation in corresponding substrates (Figure 1I). Notably, pharmacological inhibition of PRMT9 using LD2 produced effects comparable to those observed with PRMT9 KD (Figure 1J; supplemental Figure 1E).

We next transduced Molm13 VEN-R cells with a doxycycline (DOX)-inducible vector containing a puromycin resistance gene together with PRMT9 or control short hairpin RNA. PRMT9 KD was first induced by DOX, then combined with

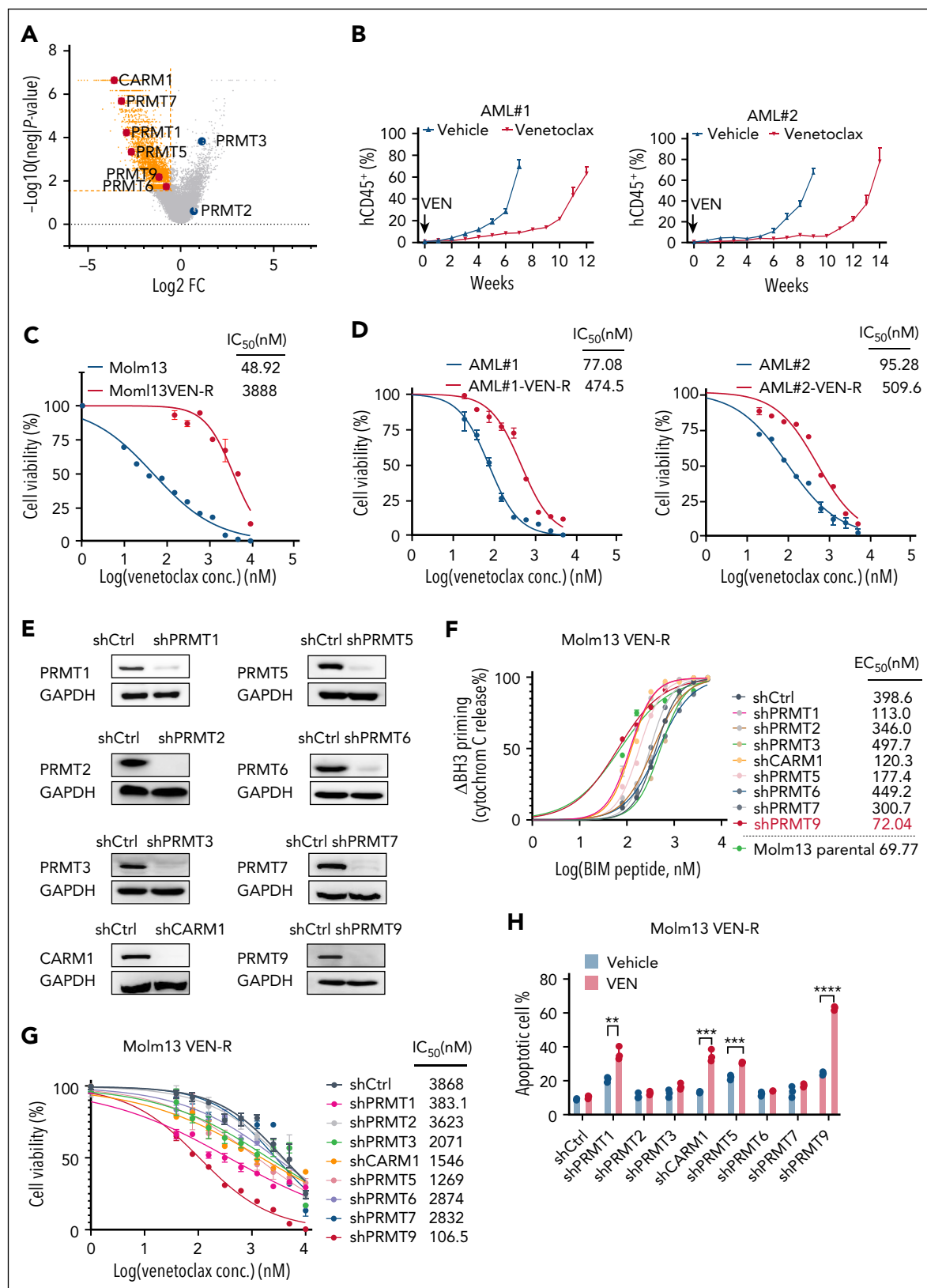


Figure 1. A loss-of-function screen shows that PRMT9 inhibition primes resistant AML cells to VEN treatment. (A) Enrichment of single guide RNAs targeting *PRMT* genes in a genome-wide CRISPR screen.²⁰ The orange area indicates positive regulators of VEN+AZA sensitivity; highlighted dots denote *PRMT* genes. (B) Percentage of circulating hCD45⁺ blasts in AML1- and AML2-engrafted mice after sequential VEN treatment. Data are presented as mean $\pm$ standard deviation (SD; $n = 3$). (C) Dose-response curves showing VEN IC_{50} values in Molm13 and Molm13 VEN-R cells after 48-hour treatment. Data are presented as mean $\pm$ SD ($n = 3$). (D) Dose-response curves showing VEN IC_{50} values in AML1-VEN-R and AML2-VEN-R cells relative to parental lines after 48-hour-treatment. Data are presented as mean $\pm$ SD ($n = 3$). (E) Western blot analysis of indicated *PRMT* family members after short hairpin RNA-mediated KD in Molm13 VEN-R cells. (F) Bcl-2 interacting mediator of cell death (BIM) peptide-based BH3 profiling of Molm13 VEN-R cells after KD of individual *PRMT*s or vehicle control. Data are presented as mean $\pm$ SD ($n = 3$). (G) Dose-response curves

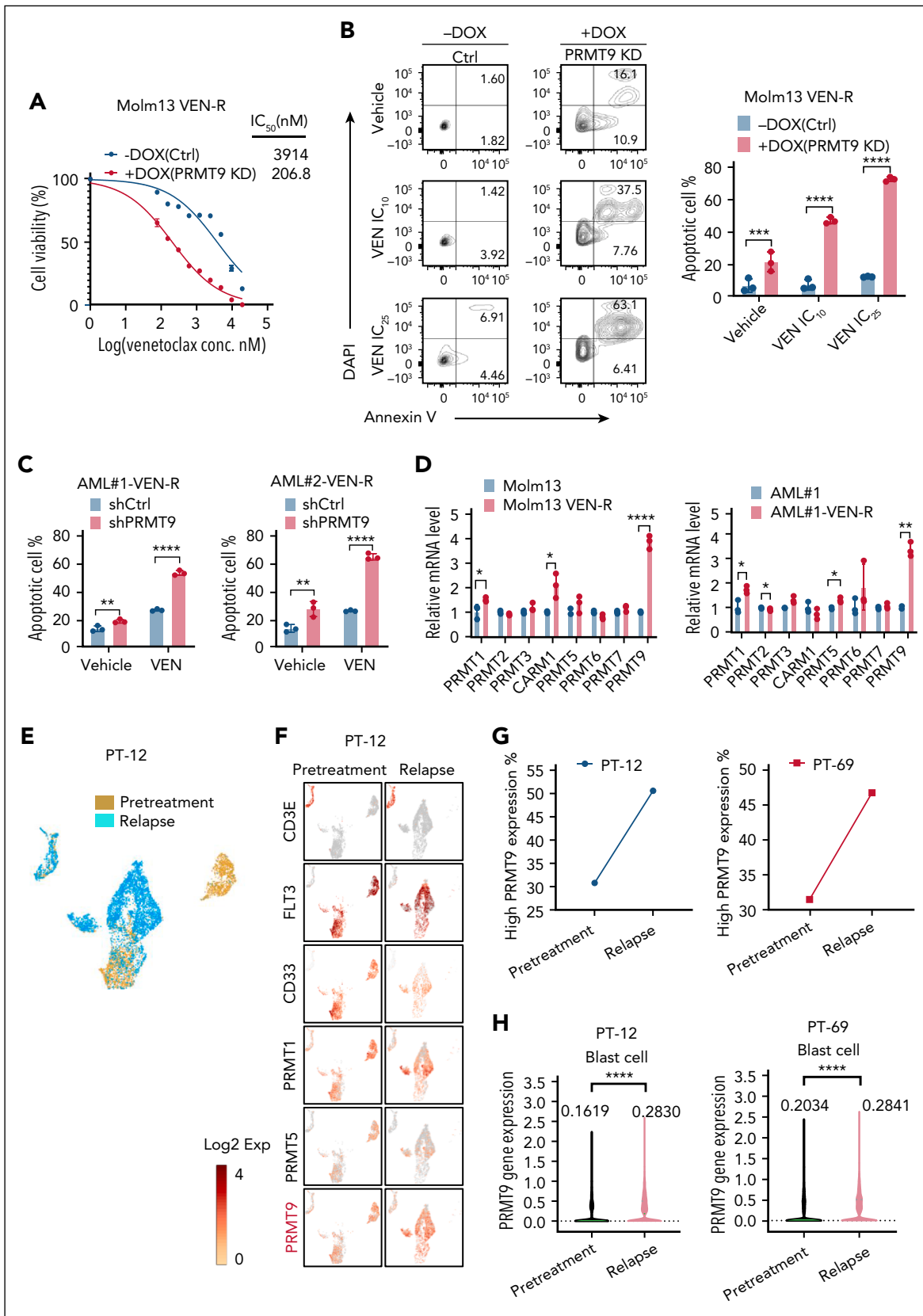


Figure 2. VEN-resistant AML cells exhibit upregulated PRMT9 expression. (A) Dose-response curves showing VEN IC_{50} values in Molm13 VEN-R cells with (–DOX) or without (+DOX) PRMT9 KD. KD was induced by 2 μ M DOX with simultaneous VEN exposure for 48 hours. Data are presented as mean $\pm$ SD ($n = 3$). (B) Schematic of flow cytometry plots of annexin V in Molm13 VEN-R cells with (–DOX) or without (+DOX) PRMT9 KD (left), apoptosis (% annexin V positive) after 2 μ M DOX-induced KD with simultaneous VEN exposure for 48 hours (IC_{10} or IC_{25}) (right). Data are presented as mean $\pm$ SD ($n = 3$). (C) Apoptosis (% annexin V positive) in AML1-VEN-R (left) and AML2-VEN-R (right) cells with or without PRMT9 KD (shPRMT9) after 48-hour VEN treatment at IC_{25} . Data are presented as mean $\pm$ SD ($n = 3$). (D) Quantitative polymerase chain

specifically elevated in relapsed blasts but not in T cells (Figure 2H; supplemental Figure 2G). In the case of PT-69, we observed uniformly high PRMT9 levels in blasts collected before treatment and after relapse (Figure 2G-H; supplemental Figure 2H-I). Notably, in both PT-12 and PT-69, most monocytic cells persisted after treatment, whereas progenitor cells were decreased at relapse (supplemental Figure 2J-K).

Genetic ablation of PRMT9 overcomes VEN resistance in vivo

To evaluate whether PRMT9 inhibition can overcome VEN resistance in vivo, Molm13 VEN-R cells engineered with inducible shPRMT9 and inducible short hairpin RNA control (ishCtrl) were implanted into NSG mice. Five days after Molm13 implantation, leukemia burden was confirmed to be equivalent across the ishCtrl and inducible shPRMT9 groups. DOX treatment was then initiated, and 4 days later, PRMT9 deletion was confirmed in Molm13 cells (supplemental Figure 3A-B). We then started 3 weeks of vehicle or VEN/AZA treatment on mice. Notably, PRMT9-KD combined with VEN/AZA significantly prolonged survival and ablated leukemia burden compared with all other groups (Figure 3A-C).

MCL1 elevation serves as a key factor in mediating VEN resistance.²⁸ To model VEN resistance in murine AML, we established an MLL-AF9(MA9)/FLT3-ITD double-hit AML transplant model by transducing BM cells from MA9 Tg mice with a lentiviral vector expressing FMS-like tyrosine kinase 3 with internal tandem duplication (FLT3-ITD). These cells showed significantly higher MCL1 and PRMT9 levels than did their single-hit MA9 counterparts. MCL1 upregulation was also seen in Molm13 VEN resistance relative to its drug-naïve counterpart (Figure 3D; supplemental Figure 9A). Notably, we observed a higher VEN IC₅₀ in double-hit cells relative to MA9 single-hit cells, which is suggestive of decreased VEN/AZA sensitivity (supplemental Figure 3C). Next, to assess the effects of PRMT9 KO in this model, MA9/FLT3-ITD double-transformed *Mx1-Cre⁺/Prmt9^{fl/fl} c-Kit⁺* cells (PRMT9-KO) or corresponding control cells were transplanted into sublethally irradiated C57BL/6 mice. Leukemia burden in both groups was detectable by bioimaging on day 7 after transplant (supplemental Figure 3D). To induce *Prmt9* deletion in hematopoietic cells, mice received polyinosinic-polycytidylic acid [poly(I:C)] injection beginning on day 7 after BMT. By day 14, PRMT9-KO was confirmed (supplemental Figure 3E). Mice were then treated for 3 weeks with either vehicle or VEN/AZA. Notably, leukemia burden was effectively ablated in the combined treatment group (Figure 3E-G).

PRMT9 inhibition disrupts RNA splicing in VEN-R AML cells

Modulation of RNA splicing reportedly enhances leukemia responsiveness to VEN.^{14,15,29} We thus asked whether PRMT9 KD may disrupt RNA splicing in VEN-R AML cells by

performing RNA sequencing in both control and PRMT9 KD Molm13 VEN-R cells and by assessing isoform use frequencies across 5 major types of alternative splicing events. Those included skipped (or retained) cassette exon, retained intron, alternative 3' splice site, alternative 5' splice site, and mutually exclusive exon.³⁰ We observed an increased number of skipped cassette exon events in PRMT9 KD relative to wild-type cells (Figure 4A; supplemental Figure 4A).

To assess a potential functional link between the aberrant splicing observed after PRMT9 inhibition and enhanced VEN sensitivity, we overlapped (1) genes exhibiting exon with alternative splicing and (2) genes reportedly showing synthetic lethality with VEN treatment.²⁰ Among overlapping genes (supplemental Table 2), only *ALG13* exhibited increased exclusion of its first coding exon in PRMT9 KD cells (Figure 4B). *ALG13*, which serves as a uridine diphosphate (UDP)-N-acetyl glucosaminyl transferase, is essential for proper N-linked glycosylation of membrane proteins.^{31,32} PRMT9 KD led to an increase in isoforms that showed skipping of the first *ALG13* coding exon (Figure 4C; supplemental Figure 4B-C) and also promoted loss of *ALG13* protein (Figure 4D; supplemental Figure 7B). Similar to treatment with N-linked glycosylation inhibitor tunicamycin, both *ALG13* KO cells and PRMT9 KD cells showed remarkable decrease in global protein N-linked glycosylation levels, based on wheat germ agglutinin staining (Figure 4E-F; supplemental Figure 4D-E), which specifically binds to N-acetylglucosamine and N-acetylneuraminic acid (sialic acid) residues.²⁹ Notably, *ALG13* KO cells showed remarkably enhanced VEN-induced apoptosis, whereas *ALG13* overexpression reduced the apoptotic sensitivity of PRMT9-KD Molm13 VEN-R cells to VEN (Figure 4G-H; supplemental Figure 4F), suggesting that *ALG13* is a downstream effector of PRMT9-KD.

Proper glycosylation is essential for membrane protein maturation.^{33,34} Thus, to understand how *ALG13* mediates VEN resistance, we assessed its function in glycosylation using the following strategy. First, we compared a list of 5572 membrane proteins (The Human Protein Atlas) with a mass spectrometry-based human glycoprotein and glycosite public database of 5722 proteins reportedly glycosylated (<http://nglycositeatlas.biomarkercenter.org/>) and identified 2074 overlapping proteins. We then refined the list by comparing it with 487 candidates identified through a published VEN resistance genetic screen²⁰ (Figure 4I). Among proteins encoded by overlapping genes, only 9 membrane proteins were reported to be linked to cancer drug resistance.³⁵⁻⁴³ Notably, our flow cytometry analysis revealed adenosine triphosphate-binding cassette subfamily C member 1 (ABCC1) to be the most significantly upregulated surface protein in Molm13 VEN-R cells relative to parental cells (Figure 4J; supplemental Figure 4G). ABCC1 protein expression was also the most significantly downregulated in *ALG13* KO Molm13 VEN-R cells (Figure 4K; supplemental Figure 4H). To assess a potential connection

Figure 2 (continued) reaction analysis of *PRMT* gene expression in Molm13 VEN-R (left) and AML1 VEN-R (right) cells relative to parental lines. Data are presented as mean ± SD (n = 3). (E) Different cell populations identified in the BM from PT-12 at pretreatment and VEN/AZA relapse stages, based on data from GSE143363. (F) Marker expression analysis in BM cells from PT-12 at pretreatment and VEN/AZA relapse stages (GSE143363). (G) Percentage of blast cells with high *PRMT9* mRNA (above mean in total cells) from PT-12 and PT-69 at pretreatment and VEN/AZA relapse stages. (H) *PRMT9* expression in blast cells from PT-12 (GSE143363; left) and PT-69 (GSE232559; right) at pretreatment and VEN/AZA relapse stages. Statistical significance was determined by 1-way ANOVA. *P < .05; **P < .01; ***P < .001; ****P < .0001. Ctrl, control; DAPI, 4',6-diamidino-2-phenylindole; shCtrl, short hairpin RNA control; shPRMT, short hairpin RNA targeting protein arginine methyltransferase.

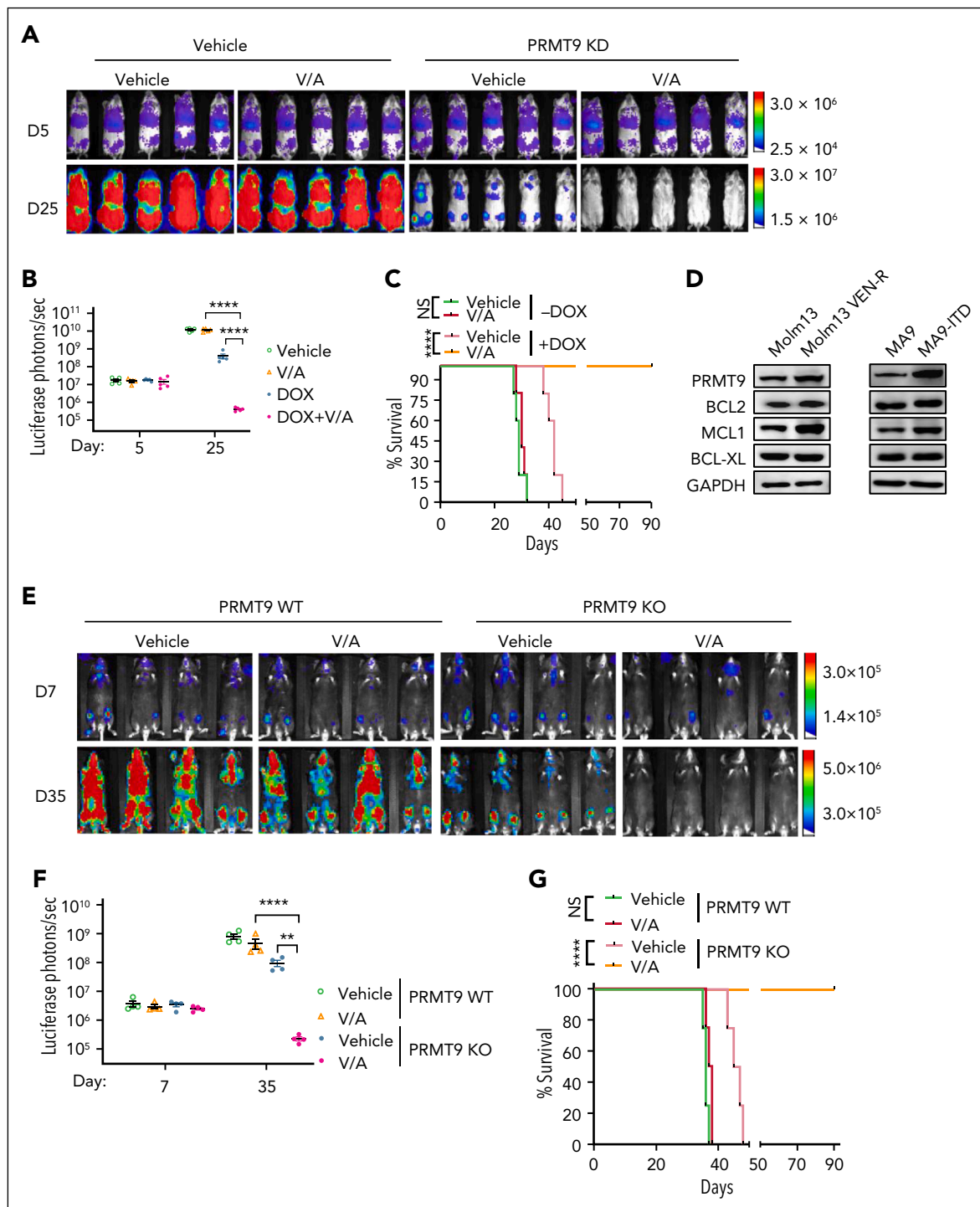


Figure 3. Targeting PRMT9 overcomes VEN resistance in mouse models. (A) Bioluminescent imaging of Molm13 VEN-R leukemia-bearing mice with inducible short hairpin RNA control (ishCtrl) or inducible short hairpin RNA targeting PRMT9 (ishPRMT9). DOX treatment started on day 5 after transplantation, followed by 3 weeks of vehicle or VEN/AZA. Leukemia burden was monitored at indicated time points in the same cohort (n = 5 per group). Color bars indicate luminescence radiance (photons/s/cm² per steradian). (B) Quantification of bioluminescent signals shown in panel A. Data are presented as mean ± SD (n = 5). (C) Kaplan-Meier survival curves of Molm13 VEN-R leukemia-bearing mice from (A). Days after transplantation are shown on the x-axis. Statistical significance was determined by the log-rank (Mantel-Cox) test. (D) Western blot analysis of PRMT9, BCL2, MCL1, and BCL-XL protein levels in Molm13 and Molm13 VEN-R cells (left) and in MA9 and MA9/FLT3-ITD cells (right). (E) Bioluminescent imaging of PRMT9 WT (wild-type; *Prmt9^{fl/f}/Mx1-Cre^{-/-}*) and PRMT9 KO (*Prmt9^{fl/f}/Mx1-Cre^{+/+}*) MA9-FLT3-ITD Luc leukemia-bearing mice. Mice received plpC and VEN/AZA treatment, and leukemia burden was assessed at the indicated time points in the same cohort (n = 4 per group). The color bar shows luminescence radiance (photons/s/cm² per steradian). (F) Quantification of bioluminescent signals shown in panel E. Data are presented as mean ± SD (n = 4). (G) Kaplan-Meier survival curves of leukemia-bearing mice from panel E. Days after transplantation are shown on the x-axis. Statistical significance was determined by the log-rank (Mantel-Cox) test. Statistical significance was determined by 1-way ANOVA. **P < .01; ****P < .0001. GAPDH, glyceraldehyde-3-phosphate dehydrogenase; NS, not significant; V/A, venetoclax (VEN) and azacitidine (AZA).

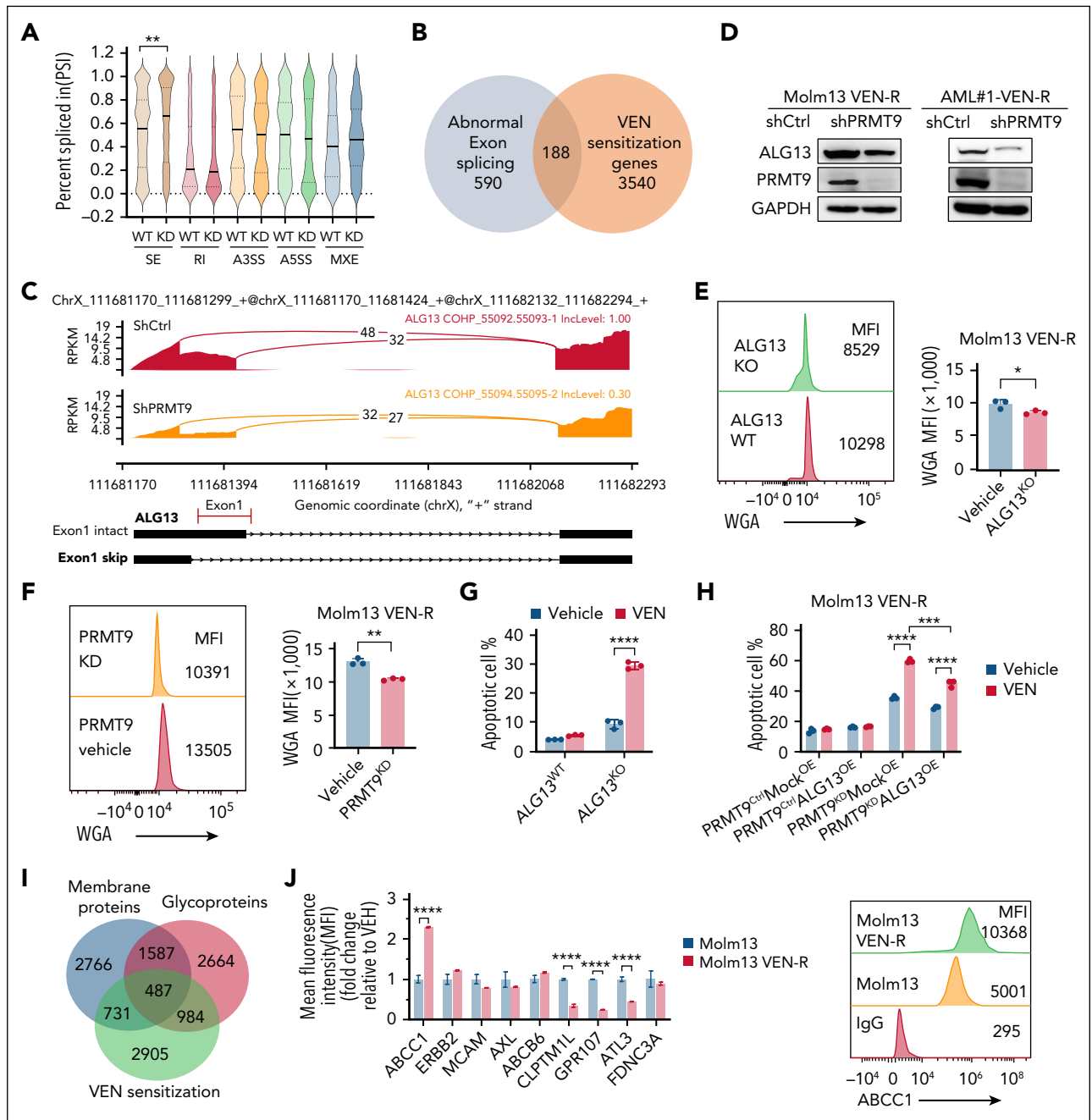


Figure 4. PRMT9 inhibition disrupts RNA splicing and suppresses translation in VEN-R AML cells. (A) Δ PSI (percent spliced in) scores representing changes in alternative splicing in PRMT9 KD AML cells relative to nontargeting control cells, based on RNA sequencing analysis. (B) Venn diagram showing overlap between exon splicing changes in PRMT9 KD AML (gray) and VEN-sensitizing genes from a CRISPR screen (orange).²⁰ (C) RNA sequencing read density and splice junction tracks showing alternative 5' splice site use at ALG13 exon 1 in WT (shCtrl) and PRMT9 KD AML cells. (D) Western blot of ALG13 and PRMT9 in Molm13 VEN-R (left) and AML1 VEN-R (right) cells after PRMT9 KD. (E) Flow cytometry (left) and quantification (right) of wheat germ agglutinin levels in Molm13 VEN-R cells with or without ALG13 KO. Data are presented as mean $\pm$ SD (n = 3). (F) Flow cytometry (left) and quantification (right) of wheat germ agglutinin levels in Molm13 VEN-R cells with or without PRMT9 KD. Data are presented as mean $\pm$ SD (n = 3). (G) Apoptosis (% annexin V positive) in Molm13 VEN-R cells with or without ALG13 KO after 48 hours of VEN treatment (IC₂₅). Data are presented as mean $\pm$ SD (n = 3). (H) Apoptosis analysis (% annexin V positive) in Molm13 VEN-R cells with or without PRMT9 KD and ALG13 overexpression after 48 hours of VEN treatment (IC₂₅). Data are presented as mean $\pm$ SD (n = 3). (I) Venn diagram showing overlap among analyzed membrane proteins, proteins in a mass spectrometry-based human glycoprotein/glycosite database, and genes identified in a CRISPR screen whose loss enhances sensitivity to VEN.²⁰ (J) Quantitative analysis (left) of levels of ABCB1, ERBB2, MCAM, AXL, ABCB6, CLPTM1L, GPR107, ATL3, and FNDC3A cell surface proteins in Molm13 and Molm13 VEN-R cells. Data are presented as mean $\pm$ SD (n = 3). Flow cytometry (right) of ABCB1 levels in Molm13 and Molm13 VEN-R cells. (K) Quantitative analysis (left) of levels of ABCB1, ERBB2, MCAM, AXL, ABCB6, CLPTM1L, GPR107, ATL3, and FNDC3A cell surface proteins in Molm13 VEN-R cell with or without ALG13 KO. Data are presented as mean $\pm$ SD (n = 3). Flow cytometry (right) of levels of ABCB1 in Molm13 VEN-R cell with or without ALG13 KO. Data are presented as mean $\pm$ SD (n = 3). (L) Flow cytometry (left) and quantification (right) of ABCB1 levels in ALG13-KO Molm13 VEN-R cells reconstituted with WT ALG13 (overexpression of wild-type) or the Δ exon 1 splice variant (overexpression of mutant). Data are presented as mean $\pm$ SD (n = 3). (M) Flow cytometry (left) and quantification (right) of ABCB1 expression in AML1-VEN-R vs parental AML1 cells. Data are presented as mean $\pm$ SD (n = 3). (N) Apoptosis (% annexin V positive) in AML1-VEN-R and Molm13 VEN-R cells treated for 48 hours with verapamil (10 μ M) or dimethyl sulfoxide in combination with VEN (IC₂₅). Data are presented as mean $\pm$ SD (n = 3). (O) Polysome profiling of RNAs from control (blue) and DOX-induced PRMT9 KD (red) Molm13 VEN-R cells after 72-hour DOX treatment. Positions of free ribosomal subunits (40S and 60S), monosomes (80S), and polysomes are indicated. (P) Quantitative

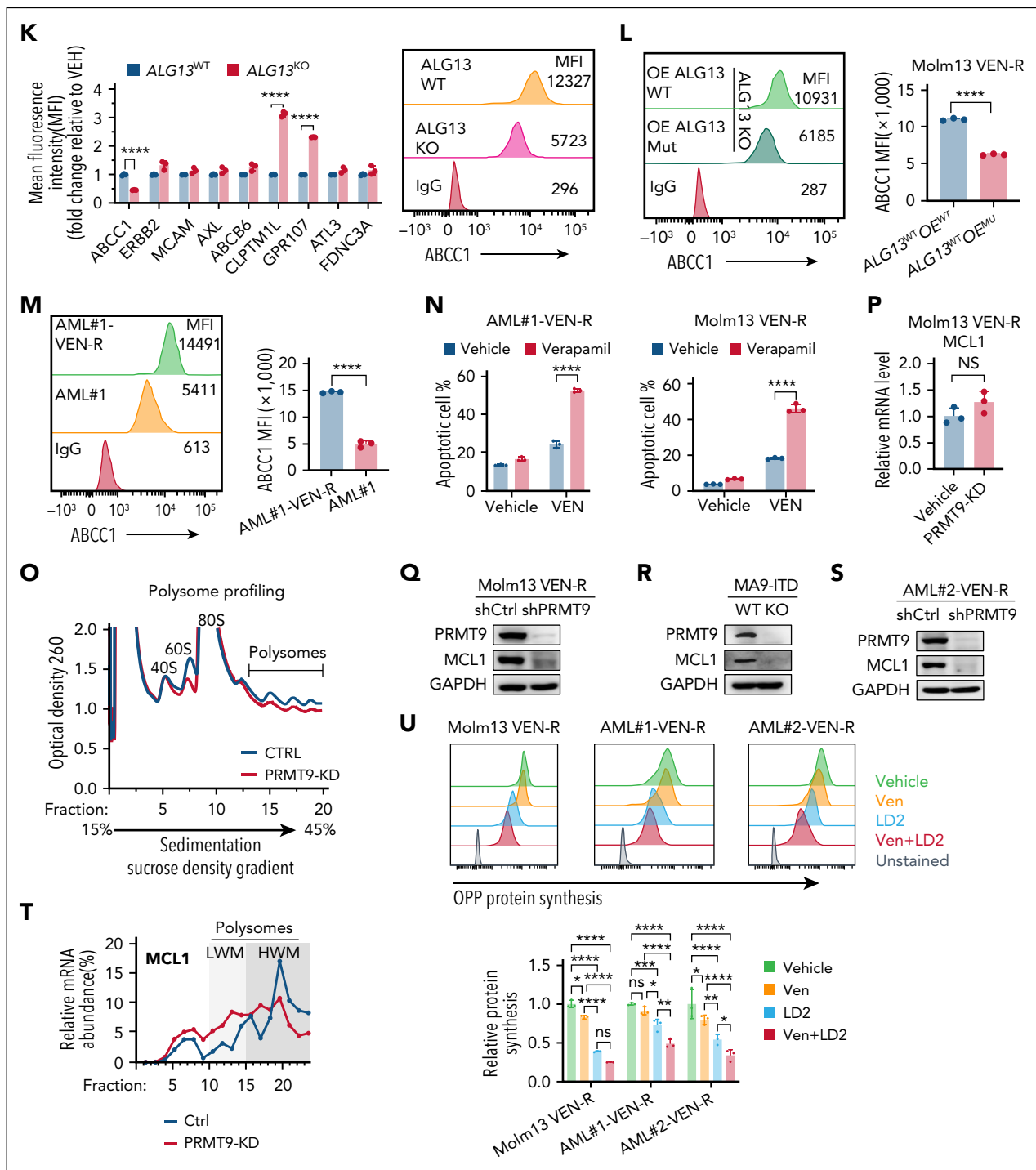


Figure 4 (continued) polymerase chain reaction analysis of *MCL1* expression in Molm13 VEN-R cells with or without PRMT9 KD. Data are presented as mean ± SD (n = 3). (Q) Western blot analysis of PRMT9 and MCL1 protein levels in Molm13 VEN-R cells with or without PRMT9 KD. (R) Western blot analysis of PRMT9 and MCL1 protein levels in MA9/FLT3-ITD cells with or without PRMT9 KD. (S) Western blot analysis of PRMT9 and MCL1 in AML2-VEN-R cells with or without PRMT9 KD. (T) *MCL1* mRNA levels in RNA isolated from the indicated polysome fractions. (U) Protein synthesis in Molm13 VEN-R, AML#1-VEN-R, and AML2-VEN-R cells treated with VEN (IC₂₅), PRMT9 inhibitor LD2 (10 μM), or their combination. The right panel shows quantification. Data are presented as mean ± SD (n = 3). Statistical significance was determined by 1-way ANOVA and unpaired 2-tailed Student t test. NS, not significant. *P < .05; **P < .01; ***P < .001; ****P < .0001. GAPDH, glyceraldehyde-3-phosphate dehydrogenase; IgG, immunoglobulin G; MFI, mean fluorescence intensity; OPP, O-propargyl-puromycin; shCtrl, short hairpin RNA control; shPRMT, short hairpin RNA targeting protein arginine methyltransferase; WGA, wheat germ agglutinin.

between ALG13 and ABCC1, we ectopically expressed either full-length wild-type or a Δ exon 1 splice variant of ALG13 in isogenic ALG13-deficient Molm13 VEN-R cells (supplemental Figure 4I). Like with ALG13 deletion, cells expressing

misspliced ALG13 mutant showed ABCC1 downregulation, an effect not seen in cells overexpressing full-length ALG13 (Figure 4L), which suggests that ALG13 missplicing leads to ABCC1 downregulation.

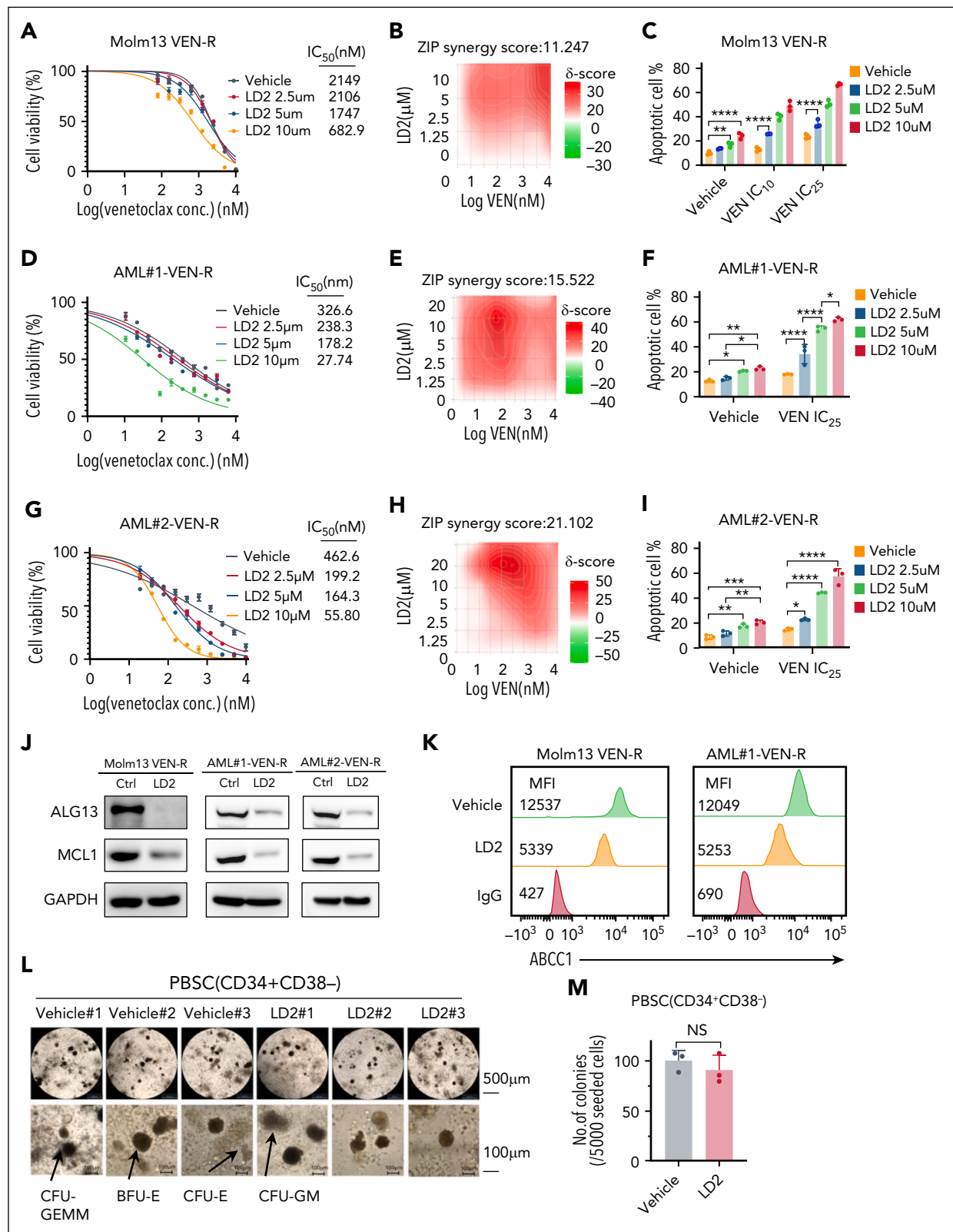


Figure 5. The PRMT9 inhibitor LD2 enhances the sensitivity of resistant cells to VEN treatment in vitro. (A) Dose-response curves of Molm13 VEN-R cells treated for 48 hours with increasing concentrations of VEN in combination with LD2. VEN IC₅₀ values are indicated. Data are presented as mean ± SD (n = 3). (B) Two-dimensional synergy plot of panel A. ZIP scores indicate synergy (positive) or antagonism (negative). (C) Apoptosis (% annexin V positive) in Molm13 VEN-R cells treated for 48 hours with VEN (IC₁₀ or IC₂₅) and/or increasing doses of LD2. Data are presented as mean ± SD (n = 3). (D) Dose-response curve of AML1-1-VEN-R cells treated for 48 hours with increasing concentrations of VEN in combination with LD2. VEN IC₅₀ values are indicated. Data are presented as mean ± SD (n = 3). (E) Two-dimensional synergy plot of panel D. ZIP scores indicate synergy (positive) or antagonism (negative). (F) Apoptosis (% annexin V positive) in AML1-1-VEN-R cells treated for 48 hours with VEN (IC₂₅) and/or increasing doses of LD2. Data are presented as mean ± SD (n = 3). (G) Dose-response curve of AML2-2-VEN-R cells treated for 48 hours with increasing concentrations of VEN in combination with LD2. VEN IC₅₀ values are indicated. Data are presented as mean ± SD (n = 3). (H) Two-dimensional synergy plot of panel G. ZIP scores indicate synergy (positive) or antagonism (negative). (I) Apoptosis (% annexin V positive) in AML2-2-VEN-R cells treated for 48 hours with VEN (IC₂₅) and/or increasing doses of LD2. Data are presented as mean ± SD (n = 3). (J) Western blot analysis of ALG13, MCL1, and GAPDH protein levels in Molm13 VEN-R, AML1-1-VEN-R, and AML2-2-VEN-R cells treated with Ctrl or LD2. (K) Flow cytometry analysis of ALG13 and MCL1 protein levels in Molm13 VEN-R and AML1-1-VEN-R cells treated with Vehicle, LD2, or IgG. (L) Colony formation assay of PBSC(CD34+CD38-) cells treated with Vehicle#1, Vehicle#2, Vehicle#3, LD2#1, LD2#2, or LD2#3. (M) Bar graph showing the number of colonies per 5000 seeded cells in PBSC(CD34+CD38-) cells treated with Vehicle or LD2. NS indicates no significant difference.

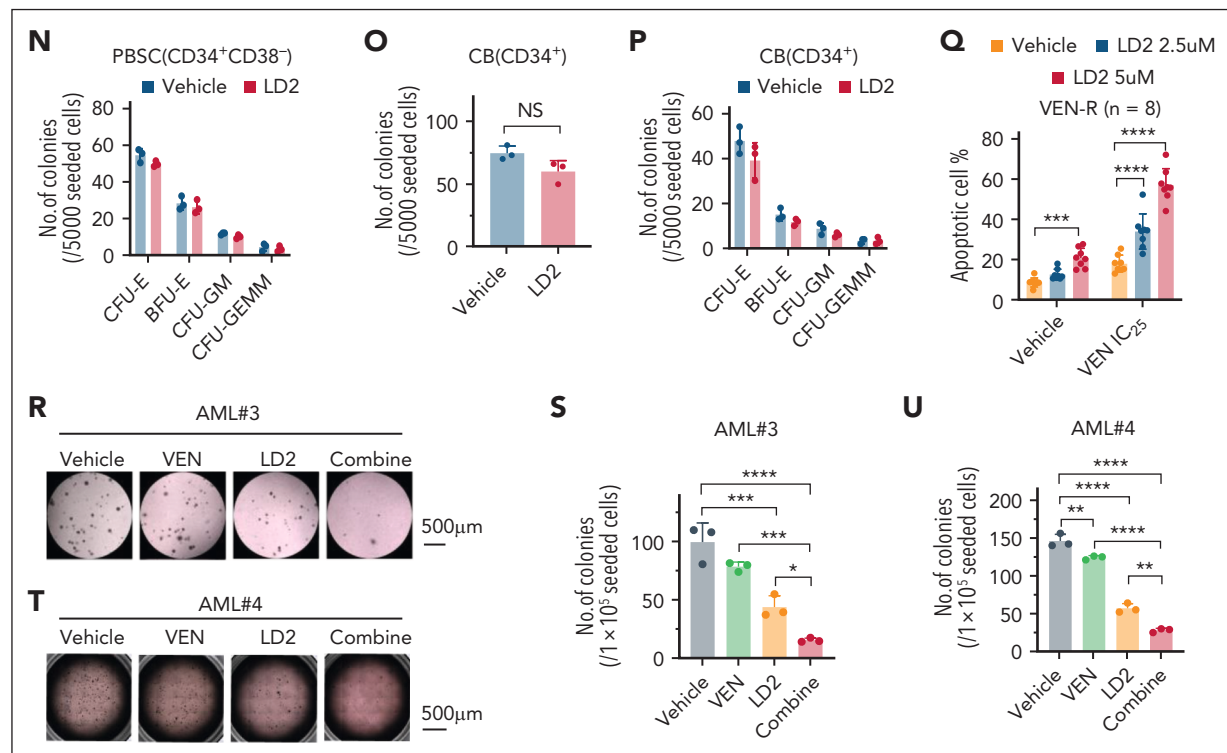


Figure 5 (continued) combination with LD2. VEN IC₅₀ values are indicated. Data are presented as mean ± SD (n = 3). (H) Two-dimensional synergy plot of panel G. ZIP scores indicate synergy (positive) or antagonism (negative). (I) Apoptosis (% annexin V positive) in AML2-VEN-R cells treated for 48 hours with VEN (IC₂₅) and/or increasing doses of LD2. Data are presented as mean ± SD (n = 3). (J) Western blot analysis of MCL1 and ALG13 protein levels in Molm13 VEN-R, AML1-VEN-R, and AML2-VEN-R cells treated with LD2 (10μM) for 48 hours. (K) Flow cytometric analysis of ABCC1 levels in Molm13 VEN-R and AML1-VEN-R cells treated with LD2 (10μM) for 48 hours. Data are presented as mean ± SD (n = 3). (L) Colony-forming units from PB stem cell CD34⁺CD38⁻ treated with LD2 (10μM) for 3 days and cultured for 14 days. Data are presented as mean ± SD (n = 3). (M) Quantification of clonal numbers from panel L. Data are shown as mean ± SD (n = 3). (N) Quantification of clonal numbers of human burst-forming erythroid progenitors, colony-forming myeloid progenitors, and multipotent progenitors shown in panel L. Data are shown as mean ± SD (n = 3). (O) Quantification of colony-forming units (CFUs) from CB CD45⁺ cells treated with LD2 (10μM) for 3 days and cultured for 14 days. Data are presented as mean ± SD (n = 3). (P) Quantification of clonal numbers of human burst-forming erythroid progenitors, colony-forming myeloid progenitors, and multipotent progenitors shown in panel O. Data are shown as mean ± SD (n = 3). (Q) Apoptosis (% annexin V positive) in VEN-R cells after 48-hour treatment with VEN (IC₂₅) and LD2 (10μM). Data are presented as mean ± SD (n = 8). (R) CFUs from VEN-R cells (AML3) treated with VEN (IC₂₅) alone, LD2 (10μM) alone, or their combination for 3 days, followed by 14 days of culture. Data are presented as mean ± SD (n = 3). (S) Quantification of clonal numbers from panel R. Data are presented as mean ± SD (n = 3). (T) CFUs from VEN-R cells (AML4) treated with VEN (IC₂₅) alone, LD2 (10μM) alone, or their combination for 3 days, followed by 14 days of culture. Data are presented as mean ± SD (n = 3). (U) Quantification of clonal numbers from panel T. Data are presented as mean ± SD (n = 3). Statistical significance was determined by 1-way ANOVA and unpaired 2-tailed Student t test. *P < .05; **P < .01; ***P < .001; ****P < .0001. BFU, burst-forming unit; Ctrl, C; CB, cord blood; IgG, immunoglobulin G; NS, not significant.

ABCC1 was recently identified as a VEN efflux transporter.³⁵ We observed ABCC1 protein elevation in AML1-VEN-R cells but not in AML2-VEN-R cells compared with the corresponding parental cells (Figure 4M; supplemental Figure 4J). ABCC1 mRNA expression was not increased in VEN-R cells (supplemental Figure 4K), which suggests posttranscriptional regulation. Analysis of the DepMap database revealed that ABCC1 protein levels are negatively correlated with VEN sensitivity (supplemental Figure 4G). Moreover, pharmacological inhibition of ABCC1 with verapamil significantly enhanced VEN-induced apoptosis in resistant models, including AML1-VEN-R and Molm13 VEN-R cells, which demonstrates synergy between ABCC1 inhibition and VEN treatment (Figure 4N). Collectively, these data suggest that PRMT9 inhibition overcomes VEN resistance by modulating ALG13 splicing, thereby downregulating ABCC1.

PRMT9 inhibition suppresses protein synthesis in VEN-R AML cells

Enhanced RNA translation is implicated in VEN resistance.⁴⁴ We thus hypothesized that inhibiting RNA translation partially

contributes to phenotypes associated with PRMT9 inhibition. To test this hypothesis, we performed polysome profiling and observed that PRMT9 KD reduces polysome-associated mRNA levels in Molm13 VEN-R cells (Figure 4O). Suppression of RNA translation reportedly downregulates levels of short-lived proteins, including MCL1.⁴⁵ Accordingly, PRMT9 downregulation markedly reduced MCL1 protein level without altering its mRNA level (Figure 4P-Q; supplemental Figure 10A). Consistently, PRMT9 inhibition also suppressed MCL1 protein levels in MA9/FLT3-ITD and AML2-VEN-R cells (Figure 4R-S; supplemental Figure 10A). Profiling analysis further revealed that PRMT9 KD shifted MCL1 transcripts from high-molecular-weight to low-molecular-weight polysomes (Figure 4T). We further used the well-established O-propargyl-puromycin assay^{46,47} to assess protein synthesis in VEN-R Molm13 cells and AML PDX cells (AML1 and AML2) after treatment with the PRMT9 inhibitor LD2. Although LD2 alone modestly reduced protein synthesis in VEN-R cells, the combination of VEN and LD2 led to a further decrease in protein translation compared with LD2 monotherapy (Figure 4V). Furthermore, treatment of VEN-R cells with HHT⁴⁸ significantly synergized with VEN, reducing cell viability and enhancing apoptosis (supplemental

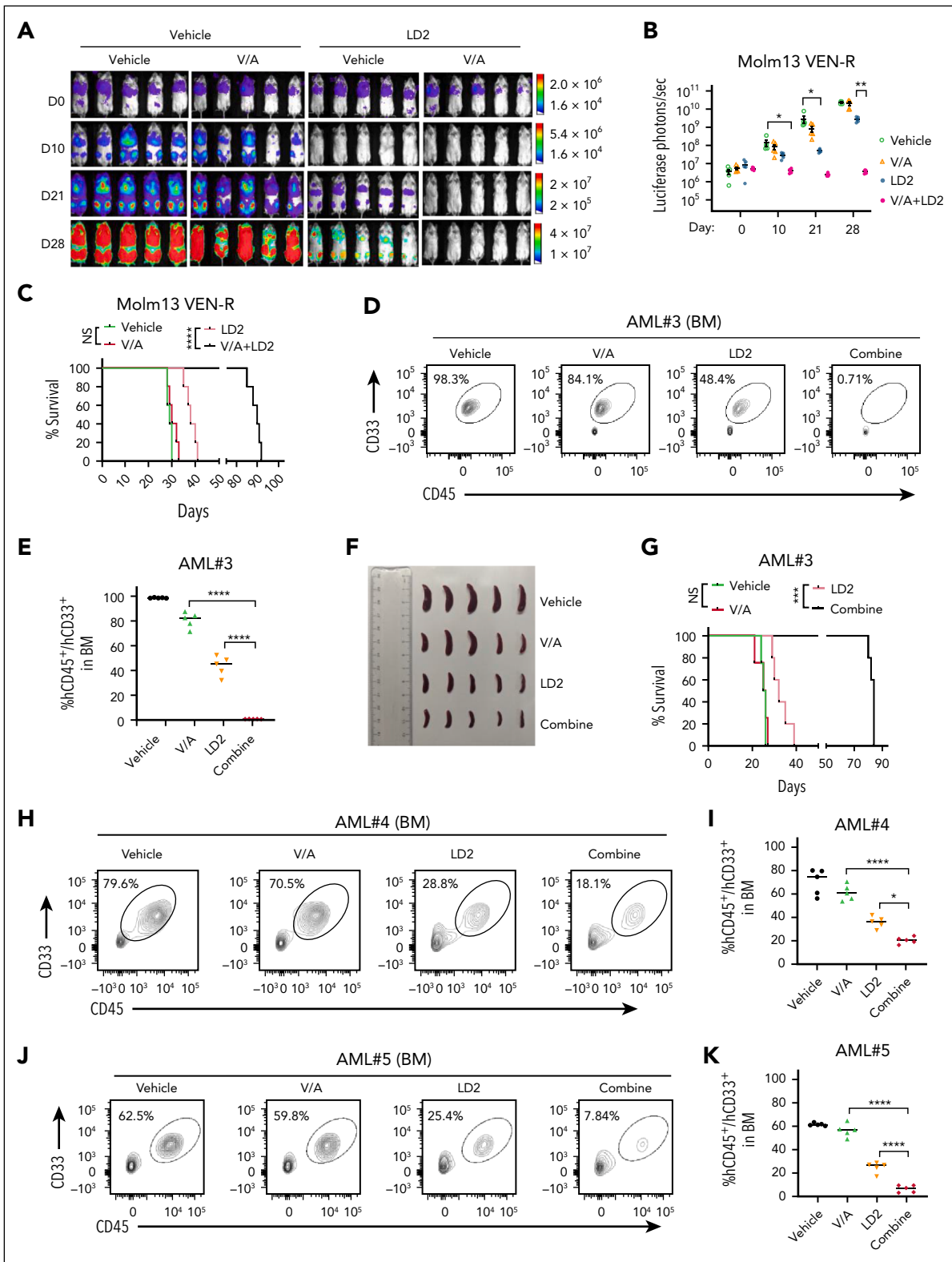


Figure 6. LD2-mediated PRMT9 inhibition potentiates VEN efficacy against resistant cells in vivo. (A) Bioluminescence imaging of recipient mice injected with Molm13 VEN-R-Luc cells. Mice were treated with LD2 (150 mg/kg, intraperitoneally, once daily) and VEN/AZA, and leukemia burden was monitored over time by bioluminescent imaging. The same cohort of mice ($n = 5$ per group) was imaged at each indicated time point. Color bar shows luminescence radiance (photons $s^{-1} cm^{-2} sr^{-1}$). (B) Quantification of bioluminescence data in panel A. Data are presented as mean $\pm$ SD ($n = 5$). (C) Kaplan-Meier survival curves of Molm13 VEN-R leukemia-bearing recipient mice from panel A. Statistical significance was determined by the log-rank (Mantel-Cox) test. (D) Schematic of flow cytometry plots showing human cell engraftment in the BM of AML3-implanted NSG mice after 3 weeks of treatment with LD2, VEN/AZA, or LD2+VEN/AZA. (E) Quantification of human cell engraftment in the BM from each treatment group shown in panel D. Data are presented as mean $\pm$ SD ($n = 5$). (F) Schematic of spleens from each treatment group shown in panel D. (G) Kaplan-Meier survival curves of AML3 leukemia-bearing recipient mice from panel D. Statistical significance was determined using the log-rank (Mantel-Cox) test. (H) Schematic of

Figure 4L-N). These findings suggest that impaired translation of mRNA also contributes to PRMT9 inhibition-induced resensitization of AML cells to VEN treatment.

Pharmacological targeting of PRMT9 enhances the sensitivity of resistant cells to VEN treatment

We next assessed the potential synergy of the PRMT9 inhibitor LD2 with VEN in AML models. The treatment of Molm13 VEN-R cells with LD2 significantly synergized with VEN to inhibit cell viability (Figure 5A-B). Although LD2 treatment modestly enhanced VEN-mediated apoptosis in Molm13 parental cells, LD2/VEN combination promoted a significantly greater apoptotic response in Molm13 VEN-R cells (Figure 5C; supplemental Figure 5A). Comparable synergy was observed in AML1-VEN-R and AML2-VEN-R cells (Figure 5D-I).

Consistent with effects seen after PRMT9 KD, LD2 treatment also downregulated MCL1, ALG13, and ABCC1 expression in AML cells (Figure 5J-K; supplemental Figure 5B; supplemental Figure 7A). LD2 treatment exhibited minimal toxicity to normal cells, as indicated by marginal reduction in colony-forming units derived from normal hematopoietic CD34⁺CD38⁻ or CD34⁺ cells (Figure 5L-P; supplemental Figure 5C). Next, we assessed LD2/VEN combination effects in primary AML cells derived from VEN-R specimens. In these samples (n = 8), VEN treatment alone induced only modest levels of apoptosis, whereas the LD2/VEN combination (at their respective VEN 25% inhibitory concentrations) significantly enhanced apoptosis relative to cells treated with VEN only (Figure 5Q). Furthermore, colony-forming assays revealed that 2 of these samples were able to form colonies; in both, the LD2/VEN combination led to a greater decrease in colony-forming units relative to VEN alone (Figure 5R-U). We also investigated whether LD2 modulated AML cell responsiveness to chemotherapy agents (AZA, cytarabine, and daunorubicin). Specifically, we treated Molm13 cells with a range of doses of each drug, with and without LD2 at a dose sufficient to inhibit PRMT9 activity (supplemental Figure 5B). Notably, LD2 treatment markedly enhanced daunorubicin-induced cytotoxicity (supplemental Figure 5D-G), consistent with a finding that daunorubicin is an ABCC1 substrate.⁴⁹

To further evaluate LD2 efficacy in promoting VEN sensitivity in vivo, we used an NSG mouse model implanted with Molm13 VEN-R-Luc cells. After confirming equal leukemia burden in recipients, grouped mice were treated for 3 weeks with vehicle, LD2 (100 mg/kg, intraperitoneally, once daily, 5 days per week), VEN/AZA, or LD2 combined with VEN/AZA. The combination significantly reduced leukemic burden compared to either monotherapy (supplemental Figure 6A-B). Furthermore, combination treatment markedly extended mouse survival compared to either monotherapy (supplemental Figure 6C). An additional cohort treated with a higher dose of LD2 (150 mg/kg, intraperitoneally, once daily, 5 days per week) in combination with VEN/AZA showed more pronounced synergistic effects, as reflected by both reduced leukemic

engraftment and improved survival (Figure 6A-C; supplemental Figure 6D). Finally, we tested the anti-AML effects of the LD2/VEN combination by using NSG mouse model implanted with 1×10^6 T-cell-depleted VEN-R AML cells (AML3). We used the same regimen as in the Molm13 VEN-R mouse model. In this setting, the combination nearly completely eliminated leukemic burden compared with LD2 or VEN/AZA alone, as determined by leukemia cell engraftment in BM and spleen (Figure 6D-G; supplemental Figure 6E-G). We performed additional in vivo PDX experiments using 2 VEN/AZA-relapsed AML samples (AML4 and AML5) and observed similar synergistic effects between LD2 and VEN (Figure 6H-K; supplemental Figure 6H-K). To evaluate the impact of LD2 on normal hematopoiesis in vivo, human cord blood CD34⁺ cells were transplanted into NSGS mice. Twelve weeks after transplant, mice were treated with LD2 (150 mg/kg) for 4 weeks. LD2 caused a marginal reduction in CD34⁺ cell numbers, with no significant effect on lineage composition or colony-forming capacity (supplemental Figure 6L-N).

Discussion

Arginine methylation, catalyzed by PRMTs, plays an important role in modulating responses to cancer therapy, including VEN responsiveness.^{19,50} Recent studies have highlighted the therapeutic potential of targeting PRMTs, such as PRMT5, to enhance the sensitivity of mantle cell lymphoma cells to VEN.¹⁹ However, a potential role for PRMTs in AML has remained unexplored. In this study, we performed a comprehensive PRMT inhibition screen and identified PRMT9 as an actionable target for sensitizing AML cells to VEN. Specifically, we found that PRMT9 inhibition promotes VEN sensitivity by reducing synthesis of the short-lived antiapoptotic protein MCL-1 and by modulating RNA splicing, leading to downregulation of ALG13 and subsequent suppression of expression of the VEN-efflux pump ABCC1. Moreover, our PRMT9 inhibitor LD2 showed remarkable synergy with VEN in eliminating AML cells.

Aberrant translation, a hallmark of cancer, drives specific changes in the cellular proteome and facilitates tumor progression and resistance to anticancer therapies.⁵¹ Indeed, some of the most common cancer-related mutations profoundly reprogram the translational machinery. For example, activating mutations in *FLT3* and genes in the *TP53* pathway are known to significantly enhance translation initiation activity.^{52,53} Moreover, targeting dysregulated translation has emerged as a promising therapeutic strategy in cancer, particularly in leukemia and in VEN-R patients. Notably, in 2012, HHT, a global protein synthesis inhibitor, became the first US Food and Drug Administration–approved translation inhibitor and is now used clinically to treat chronic myeloid leukemia and AML.⁵⁴ HHT has demonstrated synergy with VEN in inhibiting key oncogenic pathways such as MAPK/extracellular signal-regulated kinase and PI3K/AKT while simultaneously activating the p53 pathway.⁵⁵ Moreover, the HHT/VEN combination has shown encouraging results in patients with

Figure 6 (continued) flow cytometry plots showing human cell engraftment in the BM of AML4-implanted NSG mice after 3 weeks of treatment with LD2, VEN/AZA, or LD2+VEN/AZA. (I) Quantification of human cell engraftment in the BM from each treatment group shown in panel H. Data are presented as mean $\pm$ SD (n = 5). (J) Schematic of flow cytometry plots showing human cell engraftment in the BM of AML5-implanted NSG mice after 3 weeks of treatment with LD2, VEN/AZA, or LD2+VEN/AZA. (K) Quantification of human cell engraftment in the BM from each treatment group shown in panel J. Data are presented as mean $\pm$ SD (n = 5). Statistical significance was determined by 1-way ANOVA. **P* < .05; ***P* < .01; ****P* < .001; *****P* < .0001. NS, not significant; V/A, venetoclax (VEN) and azacitidine (AZA).

newly diagnosed and relapsed/refractory early T-cell precursor acute lymphoblastic leukemia.⁵⁶ In our study, we show that PRMT9 inhibition represses translation in VEN-resistant AML cells, thereby downregulating MCL-1, which is a short-lived antiapoptotic protein critical to switching cells to BCL-2 dependency. This finding is consistent with the role of MCL-1 as a key mediator of VEN resistance. Mechanistically, PRMT9 inhibition disrupts PABPC1 methylation at R493, as we previously reported, by impairing PABPC1 binding to the mRNA poly(A) tail and repressing translation.²⁶

Disruptions in RNA splicing have also been shown to sensitize cells to VEN treatment.^{14,15} A recent CRISPR-Cas9 KO screen showed that inactivation of RBM10, an RNA splicing regulator, significantly overcame VEN resistance in AML cells.¹⁵ This effect was partially mediated by promotion of XIAP missplicing, resulting in loss of its antiapoptotic function. Similarly, splicing modulation has been reported to remodel antiapoptotic dependencies in chronic lymphocytic leukemia cells, enhancing VEN sensitivity.¹⁴ In our study, RNA sequencing analysis revealed that PRMT9 inhibition induces widespread aberrations in splicing events, with a particularly strong effect on exon skipping. Here, we first overlapped aberrant splicing events with VEN sensitizer genes identified in a published KO screen and then functionally validated candidate targets. In this way, we identified *ALG13* exon 1 skipping as a key mechanism that may contribute to VEN sensitization. *ALG13*, an enzyme critical for N-linked glycosylation, is essential for the stability and maturation of various membrane proteins.^{31,57} Further investigation of membrane proteins associated with VEN sensitization revealed that *ABCC1*, a key regulator of VEN efflux,³⁵ is regulated by *ALG13*. *ABCC1* downregulation, mediated by *ALG13* missplicing, effectively overcomes VEN resistance, indicating that both are novel downstream targets of PRMT9. Although our study highlights the critical role of *ALG13* in mediating the effects of PRMT9 inhibition, splicing modulation could affect the expression of numerous proteins. Downstream effectors other than *ALG13* likely contribute to the functional outcomes observed after splicing changes induced by PRMT9 inhibition and warrant further investigation.

In conclusion, we identified a biological role for PRMT9 in mediating VEN resistance. Targeting PRMT9 with our novel compound LD2 effectively overcame VEN resistance through 2 mechanisms: modulation of RNA splicing and repression of translation. These findings not only reveal a previously unrecognized function for PRMT9 in VEN resistance but also provide a compelling rationale for the combined use of PRMT9 inhibitors and VEN as a therapeutic strategy in AML.

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Authorship

Contribution: Y.L. and X.H. conceived and designed the project; X.H., S.W., and L.L. supervised the entire project; Y.L. performed the study, analyzed data, and wrote the manuscript; X.H. designed and performed the research, analyzed data, and wrote and edited the manuscript; Lei Zhang, H.D., X.W., Y.C., and W.C. performed some experiments and provided suggestions; U.P.Y., M.L., Lianjun Zhang, S.G., G.W., and Y.-H.F. performed the research; Y.-H.K., L.C., G.M., and S.W. interpreted data and reviewed the manuscript; and L.L. initiated the topic and designed the study, analyzed and interpreted data, and wrote and edited the manuscript.

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Footnotes

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